

DEM 601 Introduction to Demographic Analysis

Exercise Set 3

TERM
Fall 2026

SET IN
lecture 3

WORTH
2% of the final grade

DUE
13 October 2026, 23:59

Readings for this lecture

Rowland, Donald T. (2003) *Demographic Methods and Concepts*
Population Change · Population Growth and Decline

Hinde, Andrew (1998) *Demographic Methods*
Population Growth

Wachter, Kenneth W. (2014) *Essential Demographic Methods*
Exponential Growth

Carmichael, Gordon A. (2016) *Fundamentals of Demographic Analysis: Concepts, Measures and Methods*
Basic Sources, Concepts, Definitions and Types of Measures

How to hand this in. By 13 October 2026, 23:59. Show your working: an answer with no arithmetic behind it earns no marks even when the number is right. **Assume throughout that all months are of equal length.** Unless a question says otherwise, use the **exponential** model and say so. Late work receives no credit.

Every table on this sheet is also a CSV in the `data/exercise03` zip file handed out with it, so no figure needs typing in by hand.

1. One rate, three models

20 marks

A small country has a population of **7,000** at mid-2014, growing at an annual rate of **50 per thousand**.

- Compute the population at mid-2024 under all three growth models: arithmetic, geometric and exponential. Give the three figures and the difference between the largest and the smallest, in people and as a percentage of the smallest.
- State in one sentence each what the three models assume about what happens between one year and the next.
- In the lecture the same three models applied to Türkiye agreed with one another to within a few hundredths of a percentage point. Here they do not. Say what is different about this population, and what that tells you about when the choice of model matters.

Per thousand. A rate quoted per thousand is a rate: divide by 1,000 before you put it in any formula.

2. Türkiye, projected to a date

26 marks

Count taken	Basis	Population
22 October 2000	census	67,803,927
31 December 2010	ABPRS	73,722,988
31 December 2017	ABPRS	80,810,525
31 December 2020	ABPRS	83,614,362
31 December 2022	ABPRS	85,279,553

The register counts on 31 December, so each register figure above is the population at the very end of the year named. The census was taken on the day shown.

- Compute the annual exponential growth rate between 31 December 2010 and 31 December 2020. State the length of the interval before you divide by it.
- Assuming that rate held, what would the population have been on 31 December 2025?
- Your answer to (b) is a projection to the same instant as a number TurkStat has already published. Look that number up, say which figure is larger and by how much, and say what the difference tells you about the assumption the question made you make.
- The first row of the table is the last traditional census. Give two reasons why a growth rate computed from that row to the 31 December 2010 row would not mean what it appears to mean.

A date is a year plus twelfths. A projection to October is not a projection to the end of the year.

3. A district, and a rate that changes

22 marks

The population of a district was **47,298** on 1 January 2002 and **54,013** at mid-2008.

- a. Compute the annual **exponential** growth rate, then the doubling time, and state the one condition under which the doubling time is a statement about the future rather than about the present.
- b. What was the population on 1 January 2005?
- c. Suppose the rate in (a) held for exactly five years after mid-2008, and that the growth rate then settled at 10 per thousand and stayed there. During which year would the district reach 100,000 people?

4. Below replacement, still growing

22 marks

Country	Children per woman	Natural increase, % a year	Population now	At replacement fertility
Niger	6.06	+3.30	26.2m	45.5m
Iran	1.70	+0.83	90.6m	105.0m
Türkiye	1.62	+0.60	87.3m	101.5m
Thailand	1.21	-0.06	71.7m	63.9m
Japan	1.21	-0.62	124.4m	92.1m

Fertility, natural increase and median age: UN World Population Prospects 2024, estimates for 2023. The last column is UN World Population Prospects 2024, momentum variant: fertility set to replacement in 2023, mortality held, no migration, run to 2100. Replacement is about 2.10 children per woman where mortality is low.

- Which of these are below replacement, and which are still growing out of their own births?
- For Iran and for Thailand, compute the **momentum ratio**: the population at replacement fertility divided by the population now. Give each as a ratio and as a percentage change.
- Two countries are on both lists in (a). In three or four sentences, how can a country have fewer children than it needs to replace itself and still grow every year with no migration? One of your two ratios in (b) is below one: say what that means, and why fertility recovering tomorrow would not prevent it.

5. The term nobody counted

10 marks

Country	Growth rate, % a year	Natural increase, % a year
Germany	+0.35	-0.37
Syrian Arab Republic	+4.92	+1.71
Ukraine	-1.55	-0.75

Fertility, natural increase and median age: UN World Population Prospects 2024, estimates for 2023.

- For each country, compute the **net migration rate** as a residual, and show the subtraction.
- One of the three grows although its deaths outnumber its births. One shrinks on both terms at once. Name them, and say in one sentence what each is doing.

THE POPULATION BLUEPRINT • CHAPTER 3

Chapter 2 measured your population once. This chapter puts it in motion.

1. Find the total population number of your chosen population for the most recent year available and for five years before that. State the source and the year for both, say whether each is a census, a register or an estimate, and say whether the basis changed between them.
2. Compute the absolute change over those five years, and then the annual growth rate under all three models. Say which one you will use for the rest of the project, and why.
3. Compute your population's doubling time at that rate. If it is shrinking, compute the halving time instead and say what changes in the interpretation.
4. Project your population five years forward at that rate. Then write one sentence naming the assumption you have just made.

Total: 100 marks